Lecture 17:
Big data and machine learning II

PPHA 34600

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From last time: big data and machine learning

Big Data

- New data sources enable us to ask new questions
- We need to think carefully about data processing
  - Raw data
  - Processed in-house
  - Outsourced processing

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Machine learning

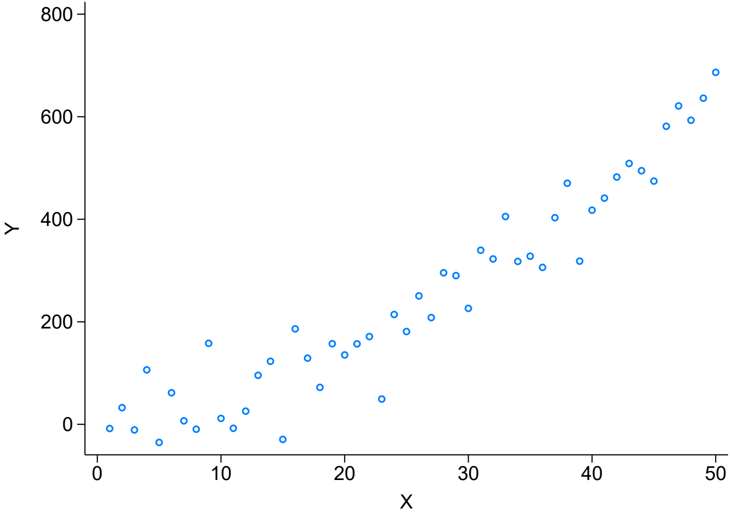
- Good at predicting $\hat{Y}$ (not at estimating $\hat{\tau}$)
- Combines in-sample fit with cross-validation

Basics of machine learning

Machine learning is just methods trying to generate predicions:

- Given a dataset with outcome Y and covariates $\mathbf{X}$, what function $f(\mathbf{X})$ best predicts Y?
- Note the difference between this and causal inference!

Predicting $\hat{Y}$



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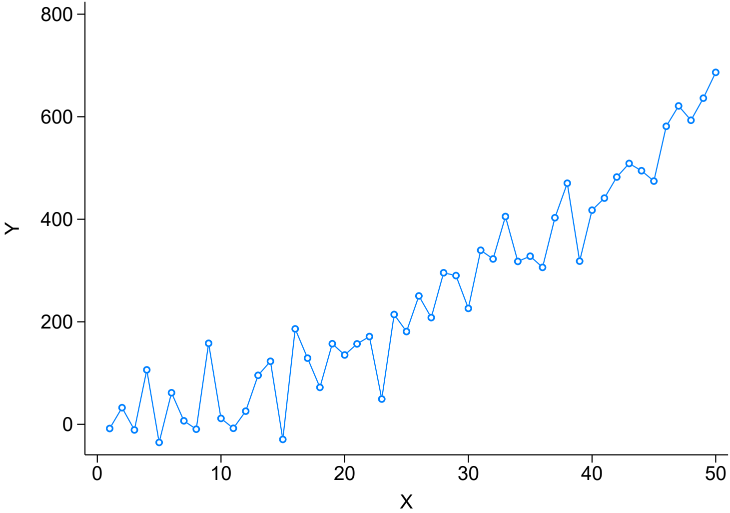

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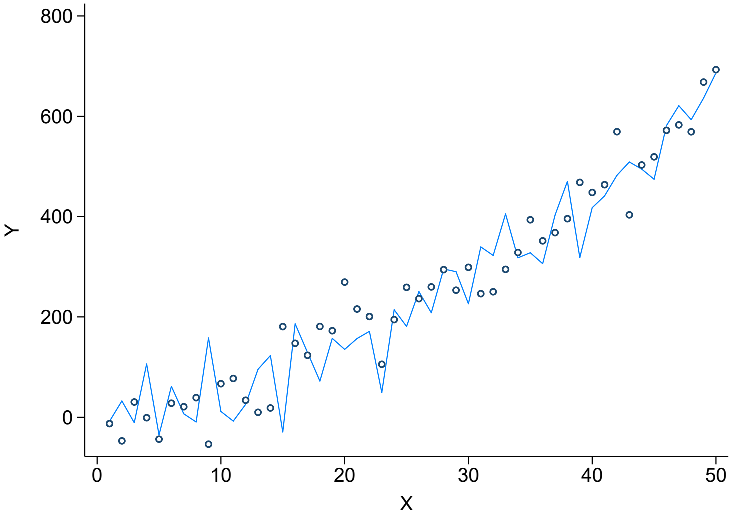

Predicting $\hat{Y}$ without overfitting

We want to come up with a good estimate of $\hat{Y}$...

... But we need to watch out for overfitting!

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   - Use an algorithm to generate the best in-sample prediction
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   - Instead of using the whole sample for step (1)...
   - Split the sample into pieces...
   - Do step (1) on one part, and predict $\hat{Y}$ on the other part
   - Record how well the model fits (eg $Y - \hat{Y}$)

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3. **Repeat**:
   - Do this several times over different sample splits
   - Pick the final model that does best

A few cautions with ML models

ML is a great tool, but we have to be careful!

- **Do NOT try to interpret the function!**
- The ML model gives you $\hat{Y}$...but *not* $\hat{\tau}$!
- We're not recovering causal effects
- And we don't get standard errors
- And the models are typically unstable

Using machine learning for causal inference

Machine learning is not designed for $\hat{\tau}$:

- We can't directly use ML for what we want to estimate
- But does this mean ML is useless for us?

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- But does this mean ML is useless for us?

$\rightarrow$ **No.**

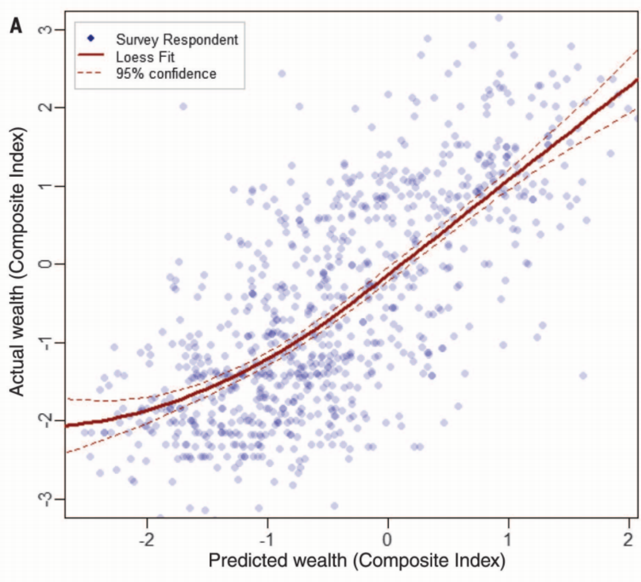
$\rightarrow$ We just need to be a little bit creative!

Using machine learning for causal inference

There are three main ways to use ML for causal inference:

1. **Data generation**
2. **Heterogeneity analysis**
3. **Estimating $\hat{\tau}$**

ML for data generation



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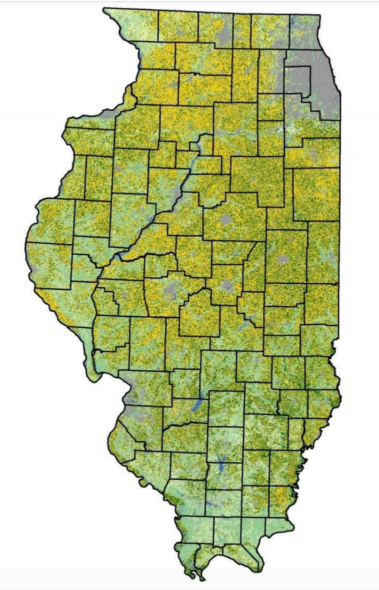

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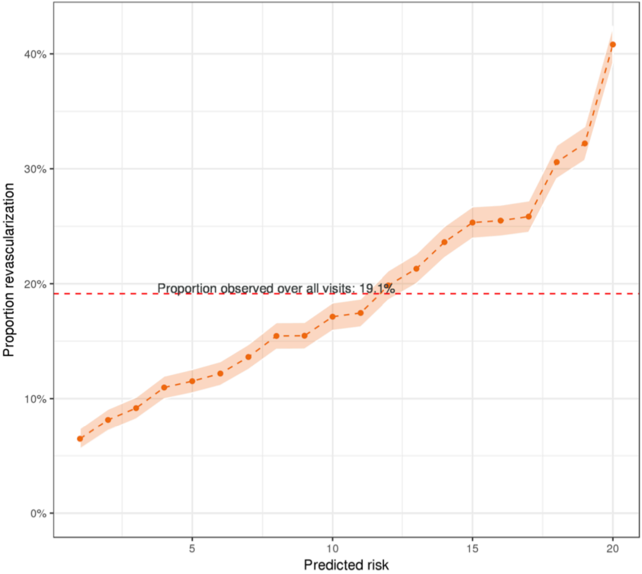

ML for heterogeneity analysis

We often want to find heterogeneous effects:

- The FPCI gets in our way for this too
- We need to compare treated vs. untreated units...
- ... *conditional* on $X_i = x$
- But there might be many X_i
- And they might even be continuous

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$\rightarrow$ Which X_i have interesting heterogeneity in $\tau_i(X_i)$?

ML for heterogeneity analysis

Reframe this into a prediction question:

- What is predicted $\hat{Y}_i(X_i)$?
- That is, which X_is give you *different* $\hat{Y}_i$?
- For this to be the same as heterogeneity in $\tau_i(X_i)$...
- ... we need random assignment to treatment
- Under random assignment, there is a 1:1 mapping between $\hat{Y}_i$ and $\hat{\tau}_i$

ML for heterogeneity analysis

The most common approach is the **causal tree**:

1. Select a training and a test sample

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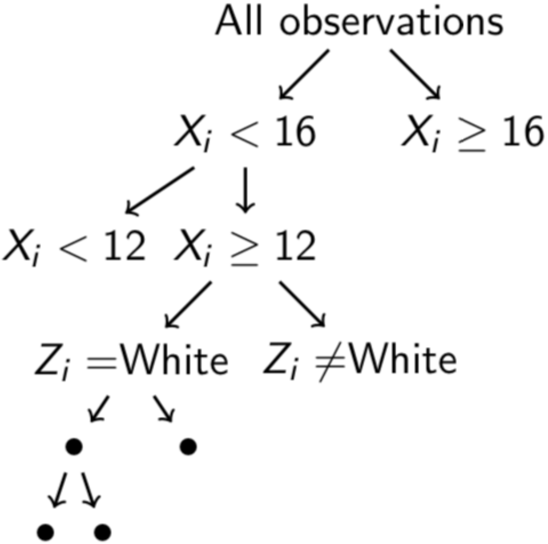
→ Repeat with different training samples to construct a causal forest

Causal trees



ML for causal inference

Now we're really pushing the frontier:

1. ML with selection on observables
2. ML with selection on unobservables

→ We need to reframe our questions as prediction problems

ML with selection on observables

Consider an underlying model:

$$Y_i = \alpha + \tau D_i + f(\mathbf{X}_i) + \varepsilon_i$$

where $E[\varepsilon|D, \mathbf{X}] = 0$: **Conditional** on $\mathbf{X}$, D is as good as random

- But which $\mathbf{X}$s matter?
- And what is the right $f(\mathbf{X})$?
- We can use ML to help us figure this out

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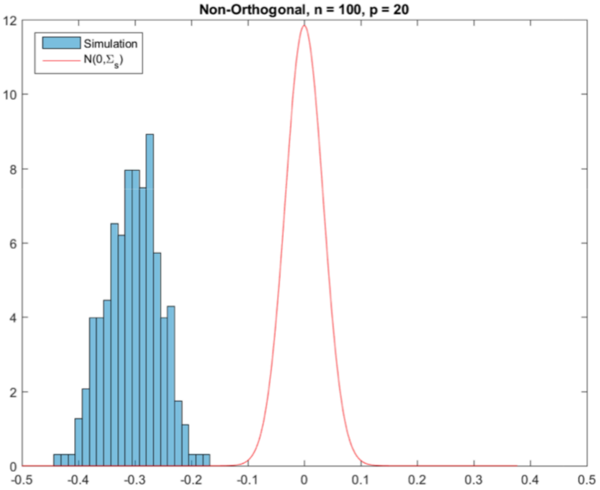
$\rightarrow$ **Simple guess:** simply predict $\hat{Y}$ based on D and $\mathbf{X}$; interpret coefficient on D as τ

ML with selection on observables



ML with selection on observables

The simple guess doesn't work!

- The overall best fit ignores the SOO assumption
- Some X_i that are important for D_i may be left out

→ ML will choose X_i that are important for Y_i

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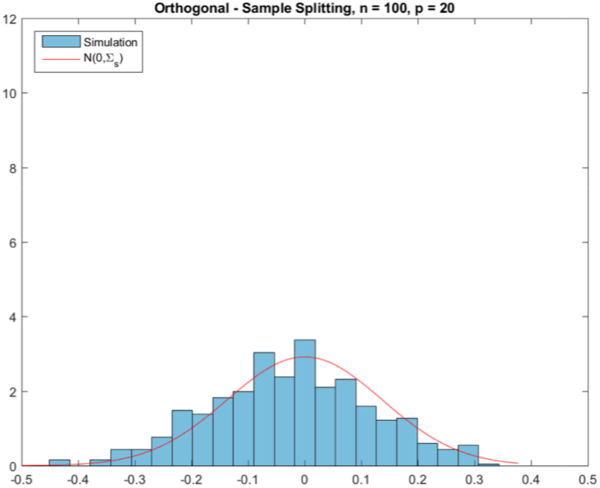
We can do better!

1. Predict $\hat{Y}$ as a function of X
2. Also predict $\hat{D}$ as a function of X
3. Estimate treatment effects using both sets of covariates

→ This only works when you do steps (1) and (2) with LASSO

ML with selection on observables



ML with selection on observables

The most up-to-date approach is:

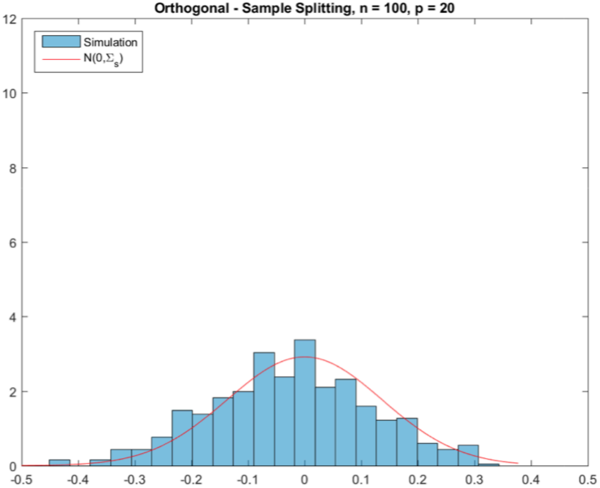
1. Predict $\hat{Y}$ as a function of X
2. Predict $\hat{D}$ as a function of X
3. Compute residuals: $Y^R = Y - \hat{Y}$ and $D^R = D - \hat{D}$
4. Recover $\hat{\tau}$ by regressing:

$$Y_i^R = \alpha + \tau D^R + \varepsilon_i$$

→ You can do steps (1) and (2) with any ML method

→ Note that this approach still needs the SOO assumptions

ML with selection on observables



ML with selection on unobservables

Just like with SOO, ML can be useful to pick covariates:

- ML helps pick X_i when we don't need them for identification
- This works for RCTs, DD, IV, etc **with exogenous covariates only**

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Just like with SOO, ML can be useful to pick covariates:

- ML helps pick X_i when we don't need them for identification
- This works for RCTs, DD, IV, etc **with exogenous covariates only**
- We can use ML when we want a better fit

Great application: first stage of IV:

- Conditional on (a) good instrument(s), we just want a good fit
- Nothing wrong with using ML to improve the first stage...
- As long as you're only using exogenous covariates

Can we use ML to build a better counterfactual?

We need to reframe as a prediction exercise:

- What is a(n estimated) counterfactual?
- Just a guess at what would've happened without treatment
- This is a simple prediction exercise
- We can potentially use ML to help us generate this counterfactual
- (Subject to all of the standard selection / identification issues)

An example: Energy efficiency in California schools

Policy issue:

- Lots of money is being spent on EE upgrades
- But are they effective?

Approach:

- Look at hourly data from 2,000 public K-12 California schools
- Some schools decided to implement EE upgrades
- This was not randomized, so we use an FE approach

$\rightarrow$ Leverage high-resolution data for an ML-augmented FE method

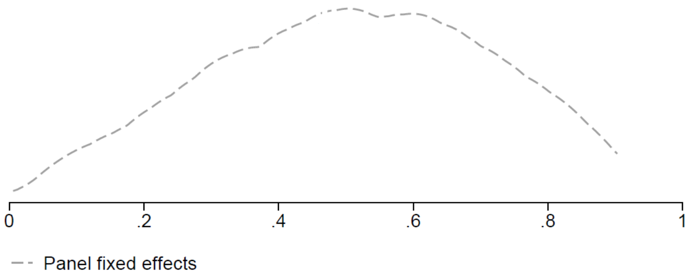
Estimating the effects of EE upgrades: version A

- We compare:
  - Consumption at schools that retrofitted to those that didn't
  - Consumption before and after retrofits
- We progressively add a series of control variables (school, hour and month-of-sample fixed effects, plus interactions):

$$Y_{ith} = \tau D_{it} + \alpha_i + \kappa_h + \gamma_t + \varepsilon_{ith}$$

Interpretation of τ: Average reduction in KWh at treated schools.

Panel FE results are unstable



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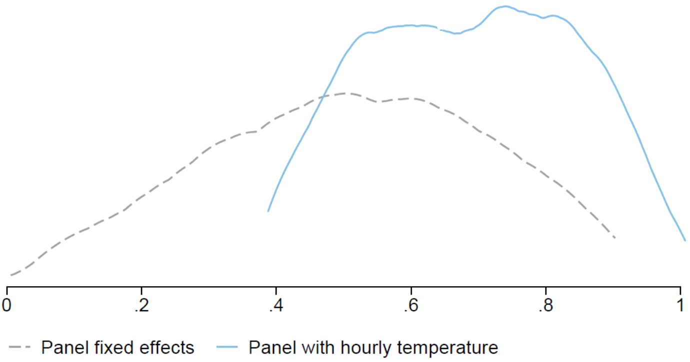

Can machine learning help?

- Panel FE models aren't properly specified.
- Schools are very heterogeneous (e.g., climate, size, school calendar).
  - Ideally, introduce school-specific coefficients and trends in a very flexible manner.
- We easily came up with $\sim$6,000,000 candidate control variables by making them school-hour specific!
- No clear *ex ante* optimal choice.

Machine Learning: Advantages in this application

- Exogenous weather variation and predictable weekly and seasonal patterns drive variation in electricity consumption.
- Schools are relatively stable consumption units:
  - as opposed to single households that move around, unobservably buy a new appliance, expand family size, etc.
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Prediction can do well!

Machine Learning: Approach

Step 1

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    - Minimizing the sum of the squared errors plus $\lambda \cdot \sum_{j=1}^{p} |\beta_j|$.
    - Larger "tuning parameters" lead to fewer coefficients.
    - Use bootstrapped cross-validation with training and holdout samples *within pre-treatment*.

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    - Larger "tuning parameters" lead to fewer coefficients.
    - Use bootstrapped cross-validation with training and holdout samples *within pre-treatment*.
  - Include a wide range of school-specific variables, and also consumption at control schools (a la synthetic control).
  - Also consider other alternatives (random forests).

Machine Learning: Approach

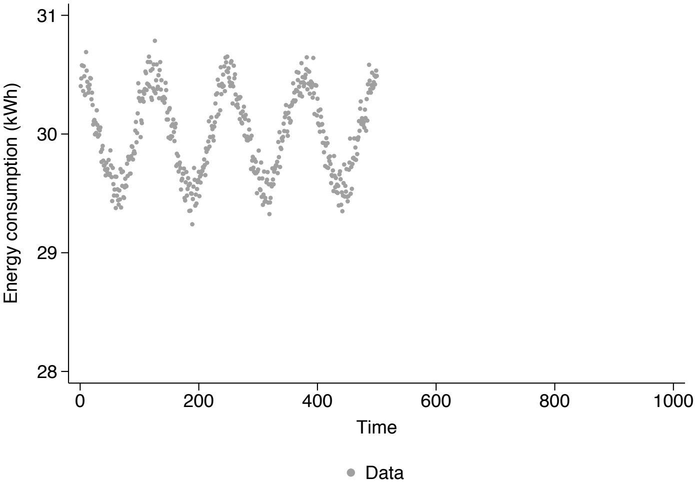
Step 2

- Regress *prediction errors* on treatment and controls.

$$Y_{ith} = \tau D_{it} + \alpha_i + \kappa_h + \gamma_t + \varepsilon_{ith}$$

  - Data pooled across schools
  - Replicates diff-in-diff approach, but Y variable is now the prediction error

Machine Learning: Graphical intuition



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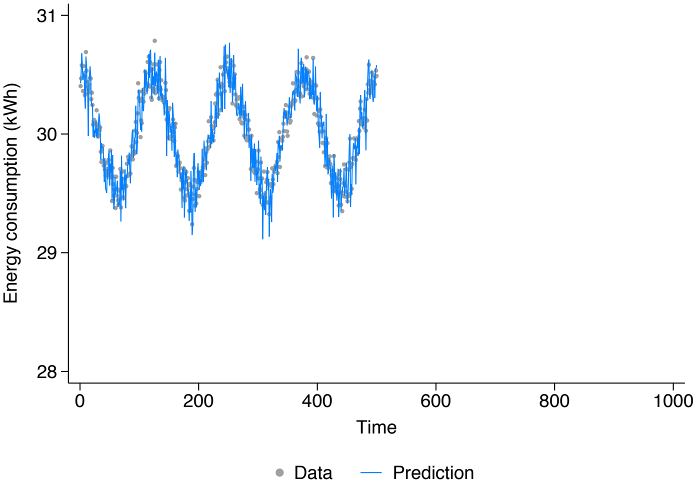

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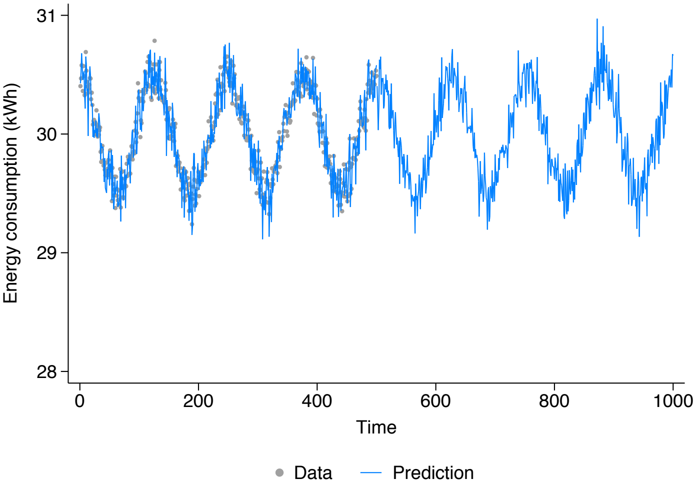

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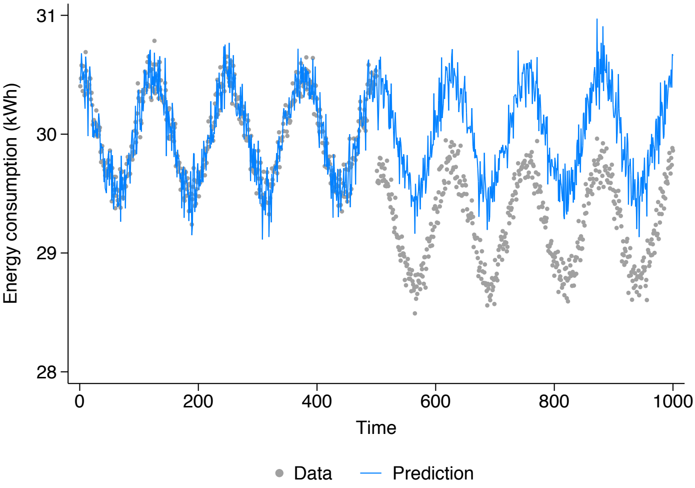
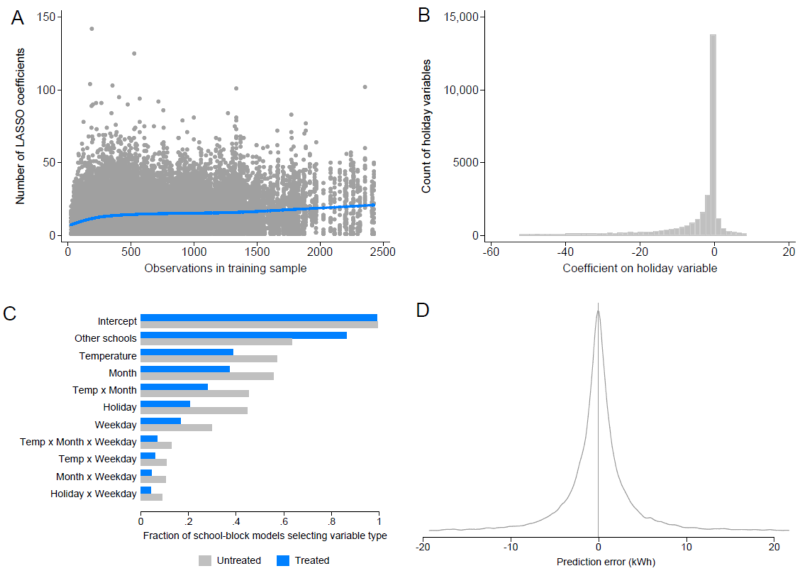
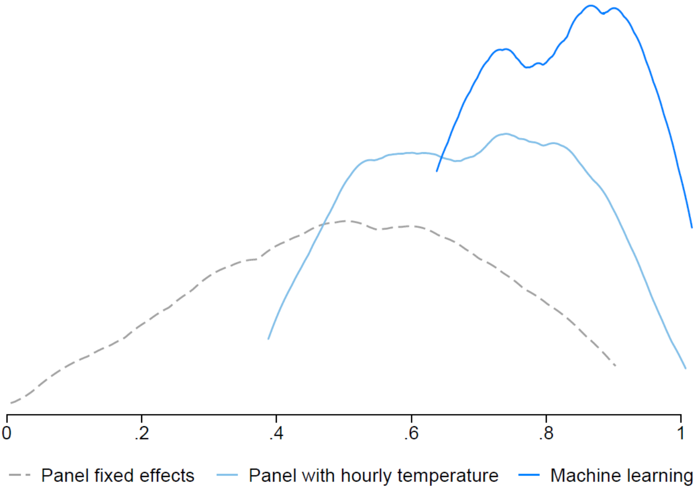


ML diagnostics



ML results are stable across estimators



Introduction to the Policy Lab

Next time:

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- **Three similar policies:** Last-mile rural electrification (with flavors)
- **Three empirical methods:** IV, RCT, RD

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$\rightarrow$ *Shift in focus*: We will spend all of class discussing the readings

Recap

TL;DR:

1. We can use ML in causal inference...
2. ...both to uncover heterogeneity
3. ...and potentially to help with identification

For next class

Topics:

- Policy lab: Does rural electrification work? I

Readings:

- Dinkelman (2011)